

COMP6237 Data Mining

Lecture 2: Discovering Groups (Clustering)

Zhiwu Huang

Zhiwu.Huang@soton.ac.uk

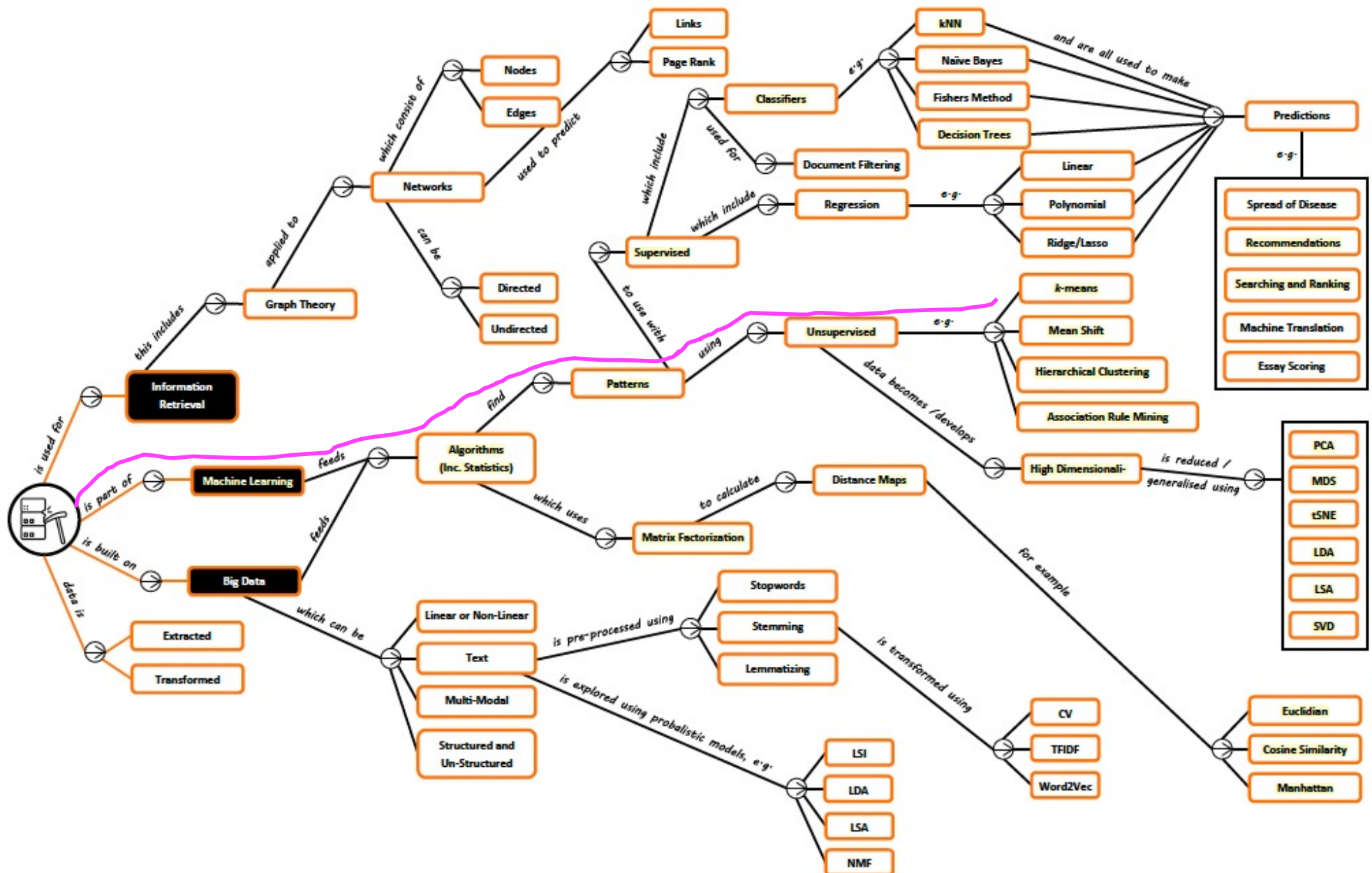
Lecturer (Assistant Professor) @ VLC of ECS
University of Southampton

Lecture slides available here:

<http://comp6237.ecs.soton.ac.uk/zh.html>

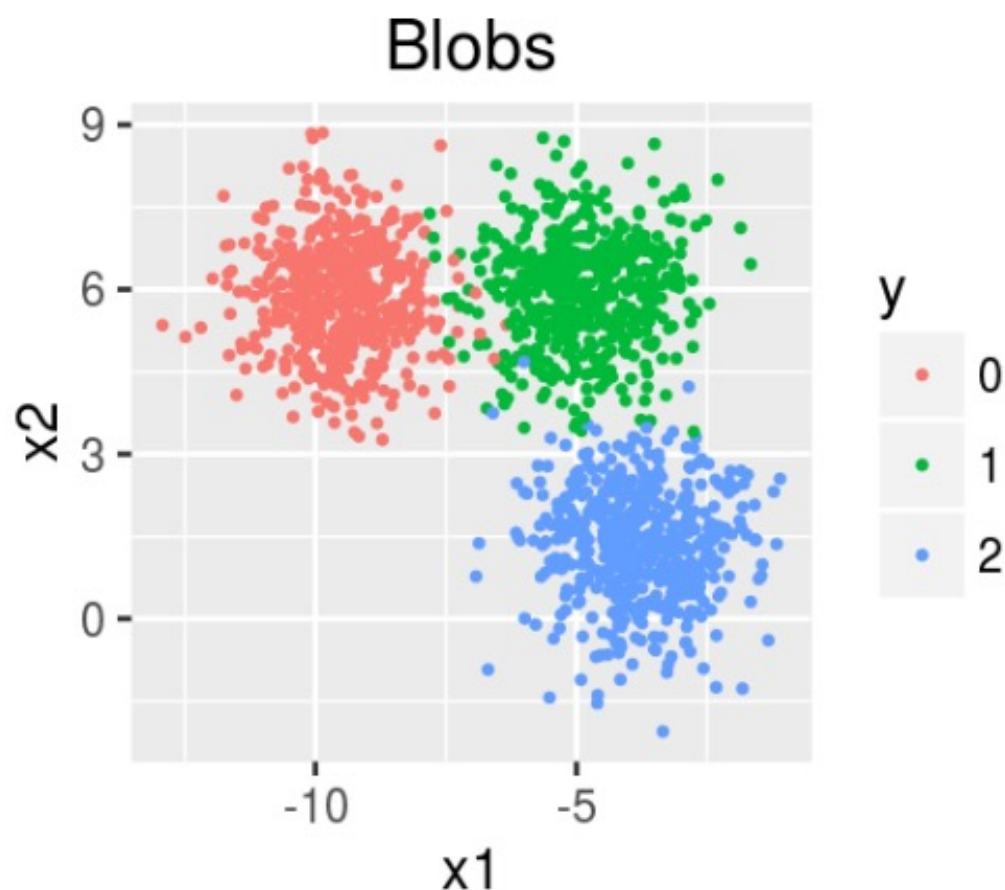
(Thanks to Prof. Jonathon Hare and Dr. Jo Grundy for providing the lecture materials used to develop the slides.)

Discovering Groups – Roadmap



Discovering Groups – Problem

- What is the clustering problem?
 - Unsupervised learning problem
 - Discover groups (i.e, clusters) in data without using any labels

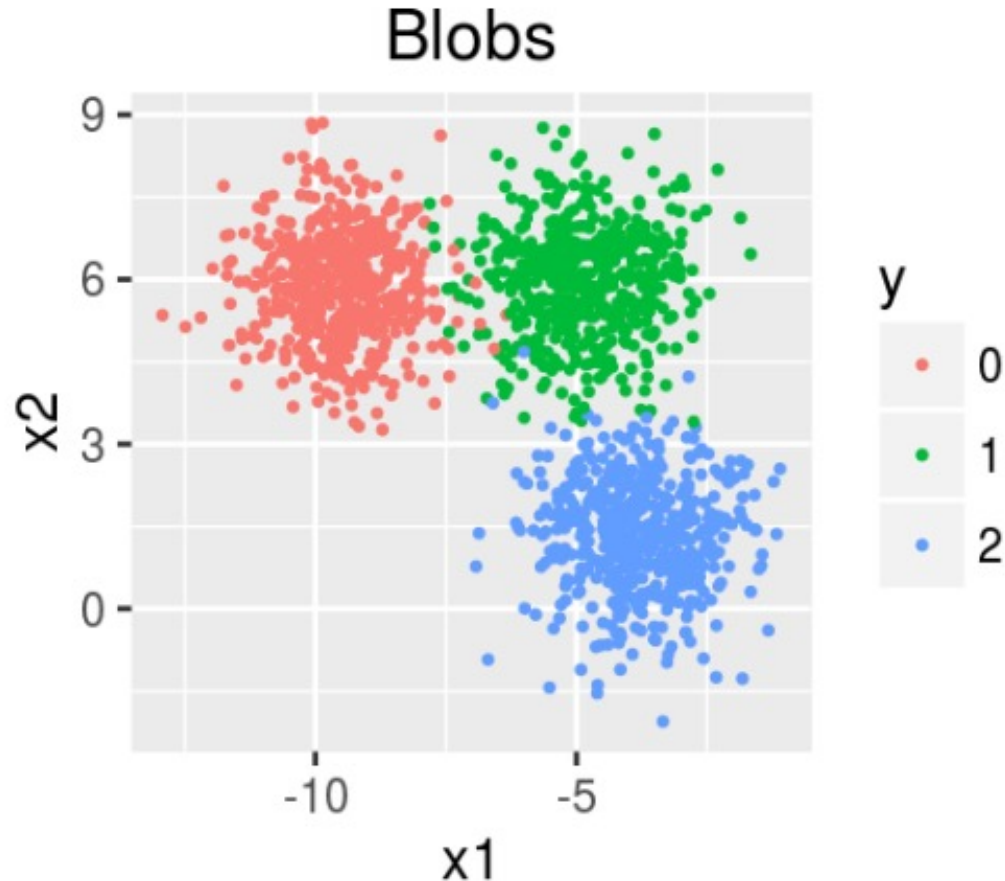


Q1: What are data points?

Q2: What are groups?

Discovering Groups – General Idea

- What is the basic idea?
 - Group data points that have similar features



Q1: What are features?

Q2: How to determine data points are similar?

Q3: But how to group automatically?

Discovering Groups – K-means

- Given: data set $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$, number of clusters K
- Goal: find cluster centers $\{\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K\}$ so that

$$J = \sum_{n=1}^N \sum_{k=1}^K r_{nk} \|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2$$

is minimal, where $r_{nk} = 1$ if $\mathbf{x}_n$ is assigned to $\boldsymbol{\mu}_k$

- Idea: compute r_{nk} and $\boldsymbol{\mu}_k$ iteratively Otherwise, $r_{n,k} = 0$

The used objective function is **Sum of Squared Error (SSE)**

Discovering Groups – K-means

Algorithm 1: K Means clustering

Data: X , K

initialise K centroids;

while *positions of centroids change* **do**

for *each data point* **do**

 | assign to nearest centroid

end

for *each centroid* **do**

 | move to average of assigned data points

end

end

return centroids, assignments;

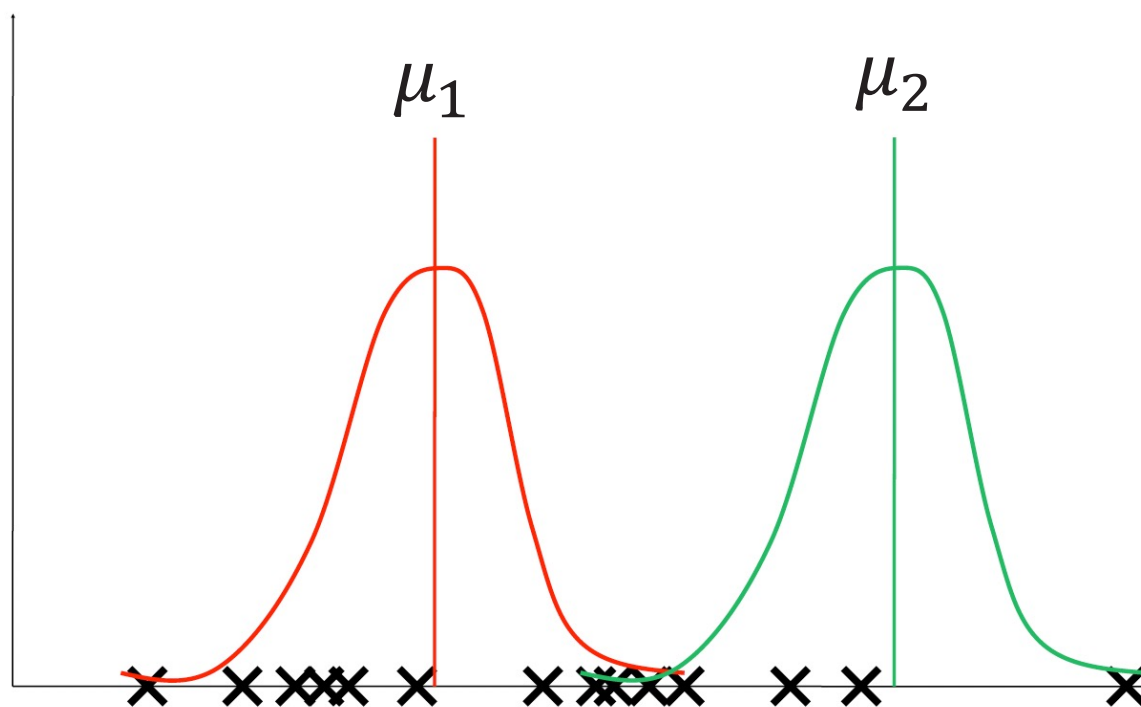
A special case of Expectation Maximisation - why?

Assumes spherical clusters

Discovering Groups – K-means

1D Example

Initialize cluster means: $\{\mu_1, \dots, \mu_K\}$

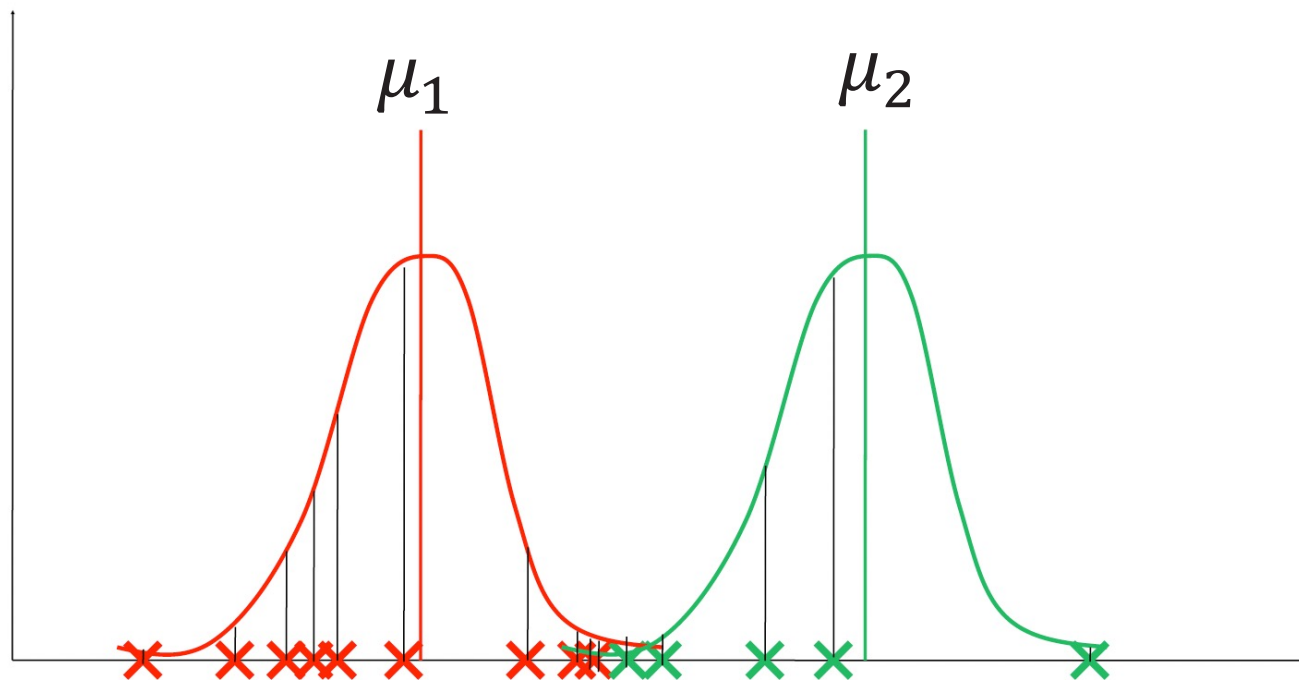


Discovering Groups – K-means

1D Example

Find optimal assignments:

$$r_{nk} = \begin{cases} 1 & \text{if } k = \arg \min_j \|\mathbf{x}_n - \boldsymbol{\mu}_j\| \\ 0 & \text{otherwise} \end{cases}$$



Source: D. Cremers

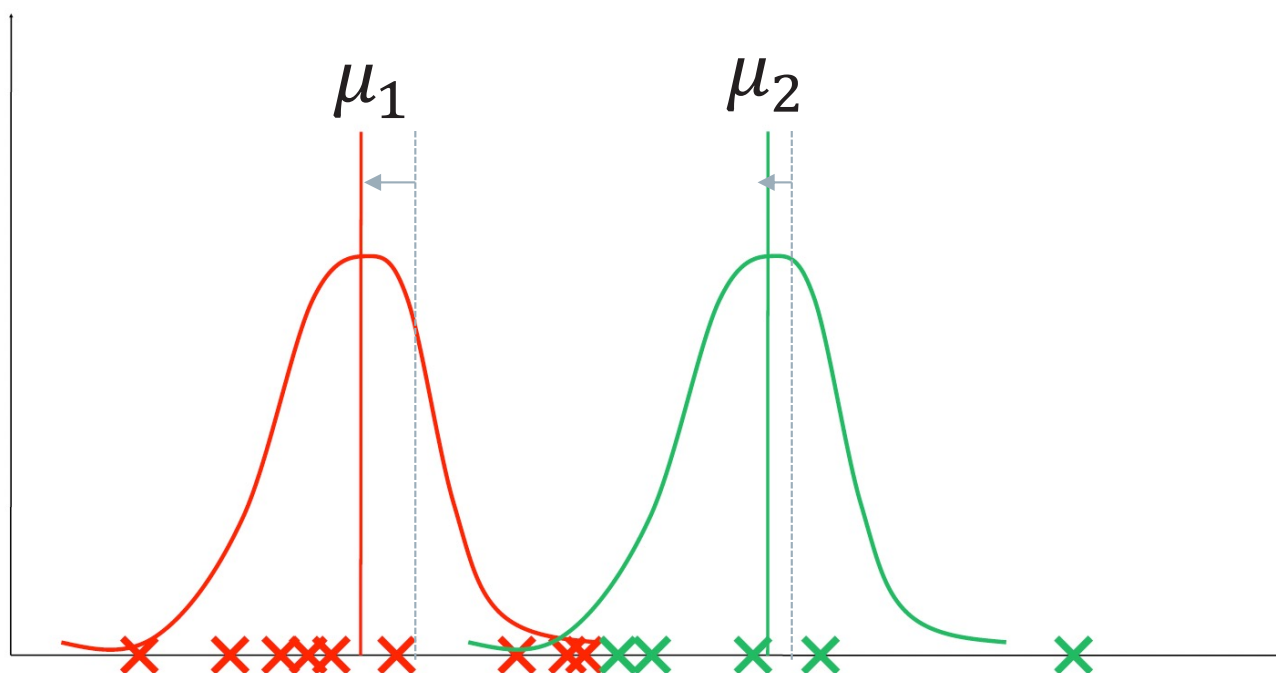
Discovering Groups – K-means

1D Example

Find new optimal means:

$$\frac{\partial J}{\partial \mu_k} = 2 \sum_{n=1}^N r_{nk} (\mathbf{x}_n - \mu_k) \stackrel{!}{=} 0$$

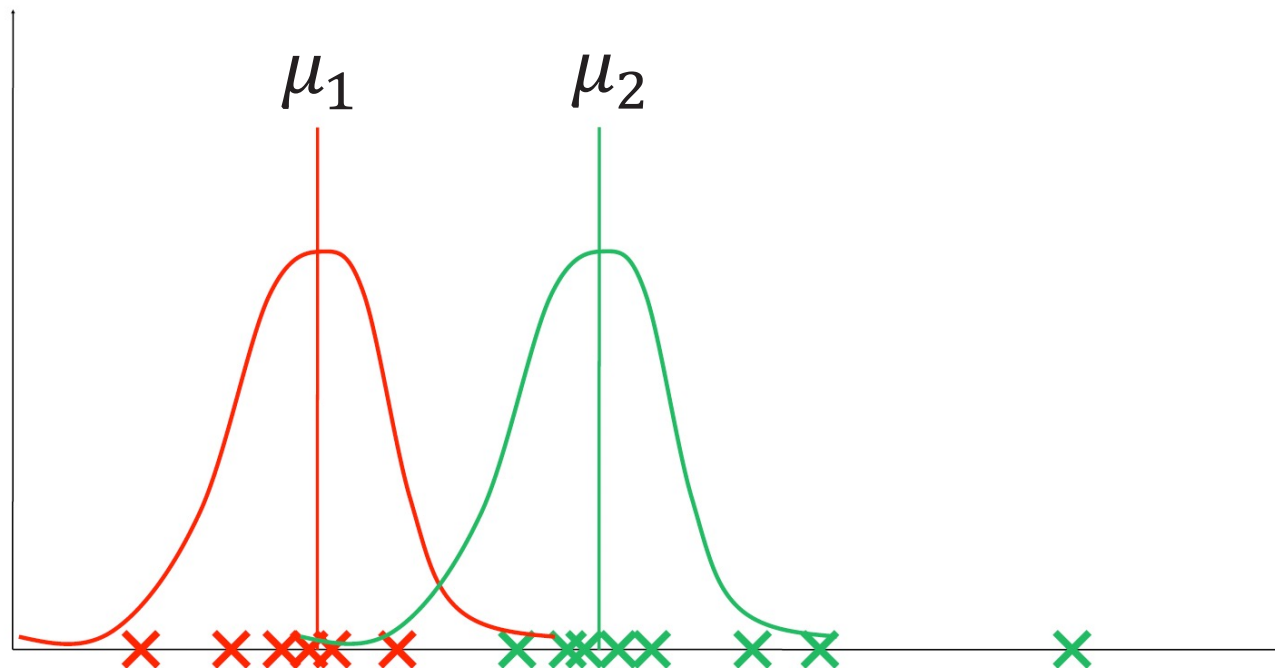
$$\Rightarrow \mu_k = \frac{\sum_{n=1}^N r_{nk} \mathbf{x}_n}{\sum_{n=1}^N r_{nk}}$$



Discovering Groups – K-means

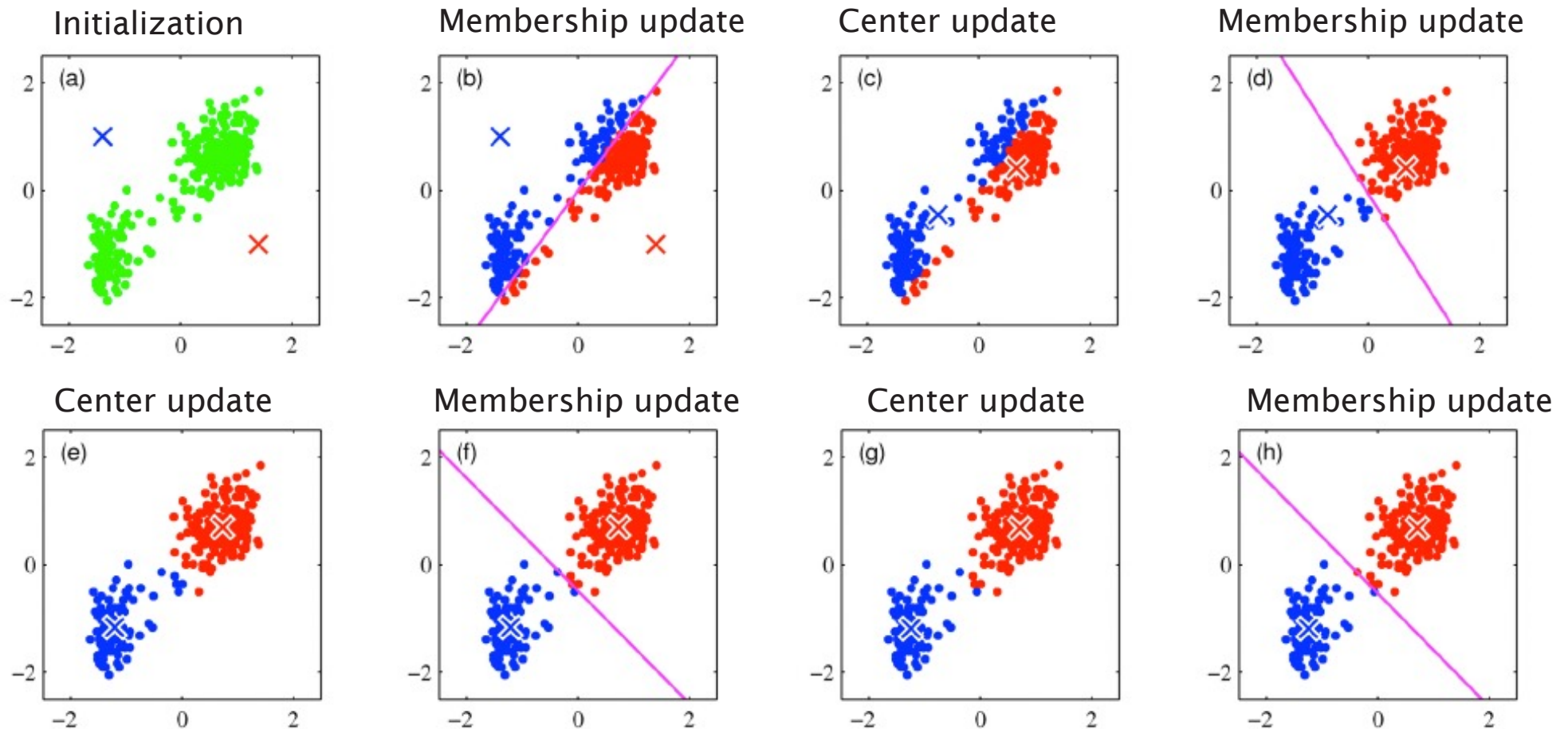
1D Example

Iterate these steps until means and assignments do not change any more



Discovering Groups – K-means

2D Example



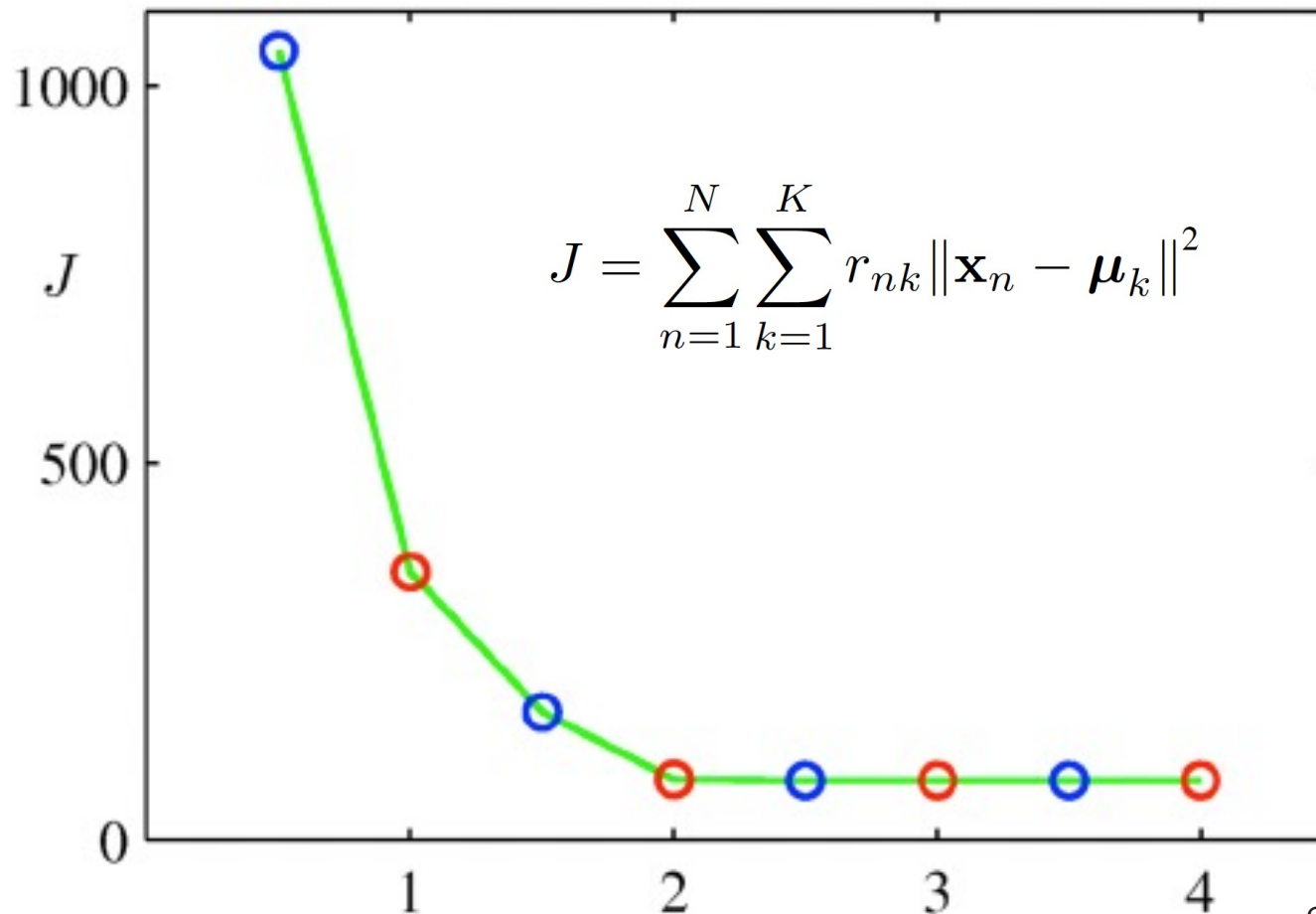
Source: D. Cremers

- Real data set
- Random initialization

- Magenta line is 'decision boundary'

Discovering Groups – K-means

Sum of Squared Error (SSE) Curve



Source: D. Cremers

- After every step the cost function J is minimized
- Blue steps: update assignments
- Red steps: update means
- Convergence after 4 rounds

Discovering Groups – K-means

K-means can quickly and cheaply cluster data.

Problems?

- need to specify the cluster number k
- depends on good initial centroid guesses
- may converge on local minimum
- assumes spherical data (or ellipsoid-shaped clusters, or at best convex clusters)

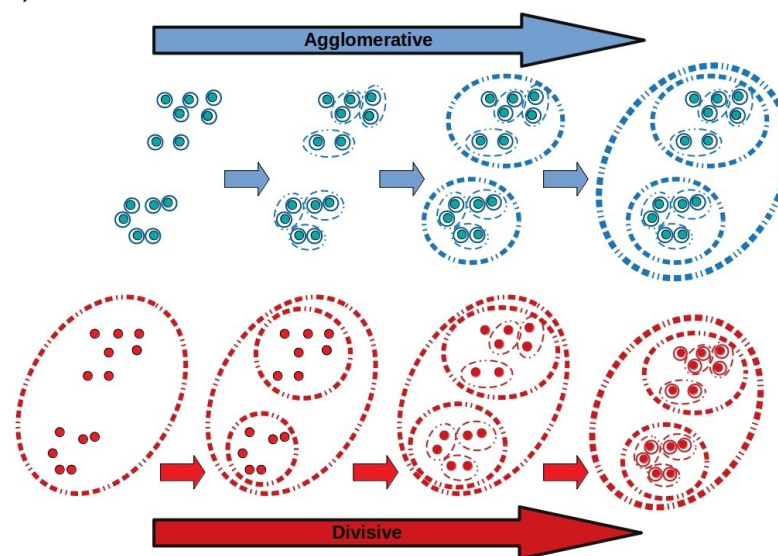
Gaussian Mixture models (GMM) can work better, using a generalization of K-means (assuming each cluster is Gaussian), not discussed in this lecture.

Discovering Groups – Hierarchical Clustering

Creates a **binary tree** that **recursively groups** pairs of similar items or clusters

Can be:

- Agglomerative (bottom up)
- Divisive (top down)



We will look at Agglomerative clustering. Needs a distance measure.

Discovering Groups – Hierarchical Clustering

Distance based linkage:

- ▶ **Minimum, or single-linkage clustering** Distance between two closest members

$$\min d(a, b) : a \in A, b \in B$$

Produces long, thin clusters

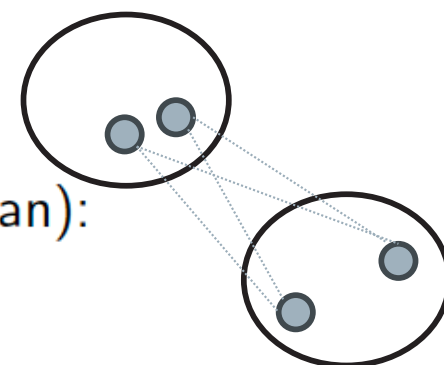
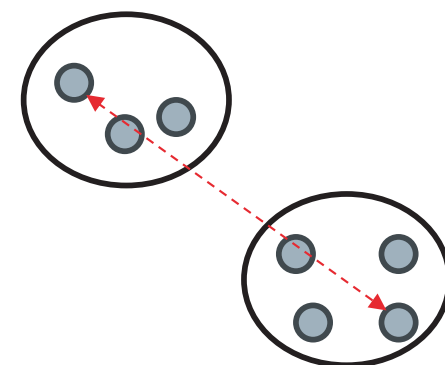
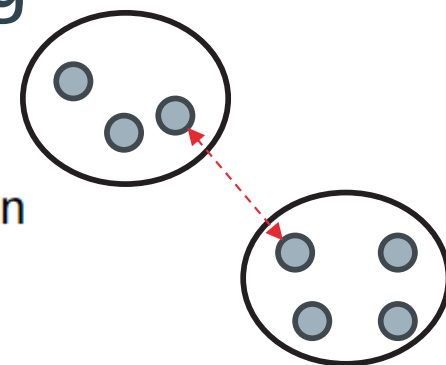
- ▶ **Maximum, or complete-linkage clustering** Distance between two most distant members

$$\max d(a, b) : a \in A, b \in B$$

Finds compact clusters, approximately equal diameter

- ▶ **Mean or Average Linkage Clustering (UPGMA:**
Unweighted Pairwise Group Method with Arithmetic Mean):

$$\frac{1}{|A||B|} \sum_{a \in A} \sum_{b \in B} d(a, b)$$



Discovering Groups – Agglomerative Clustering

Algorithm 4: Hierarchical Agglomerative Clustering

Data: N data points with feature vectors X_i $i = 1 \dots N$

$numClusters = N$;

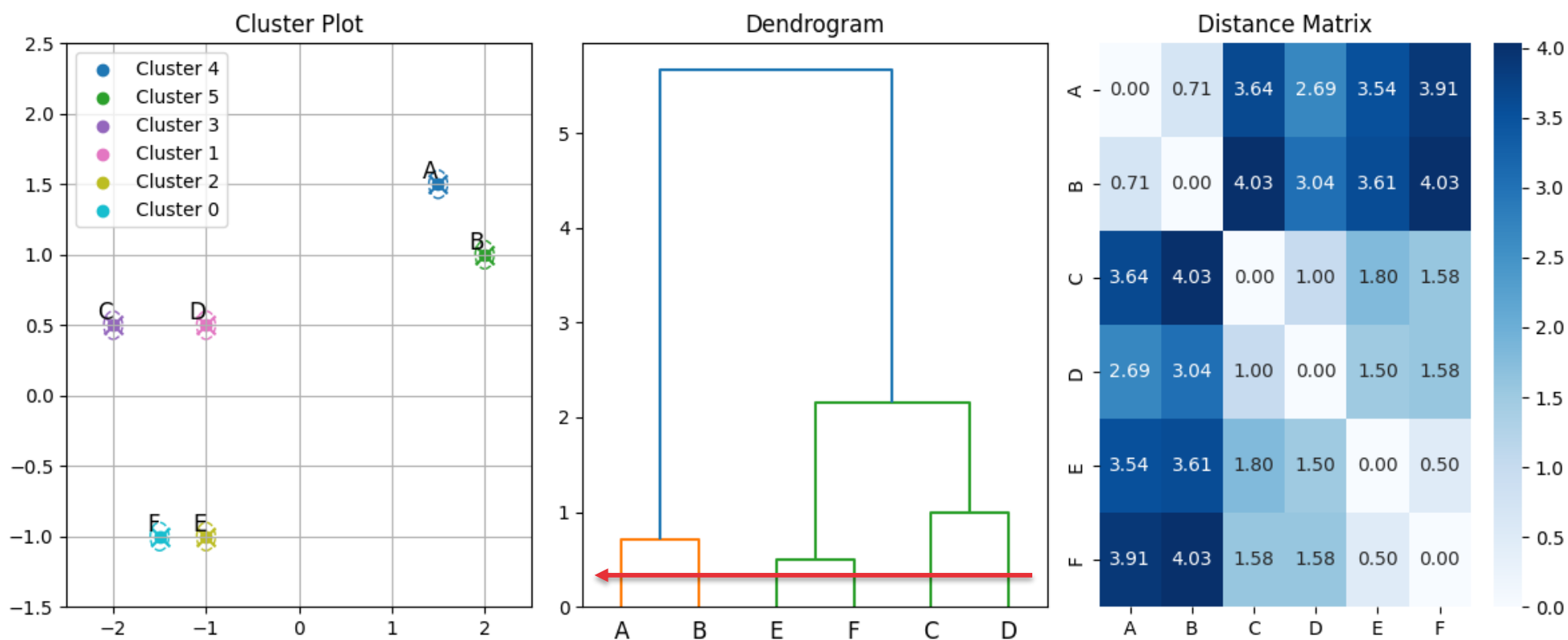
while $numClusters > 1$ **do**

 cluster1, cluster2 = FindClosestClusters();

 merge(cluster1, cluster2);

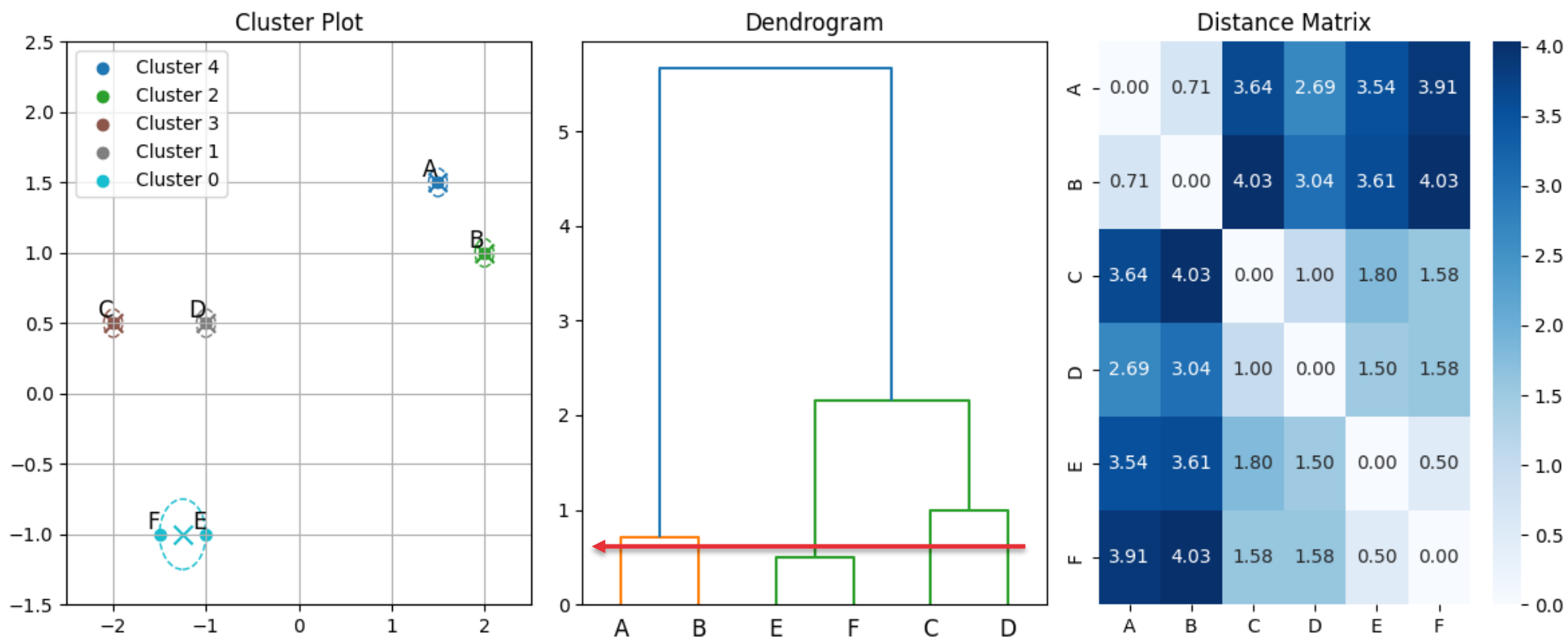
end

Discovering Groups – Agglomerative Clustering



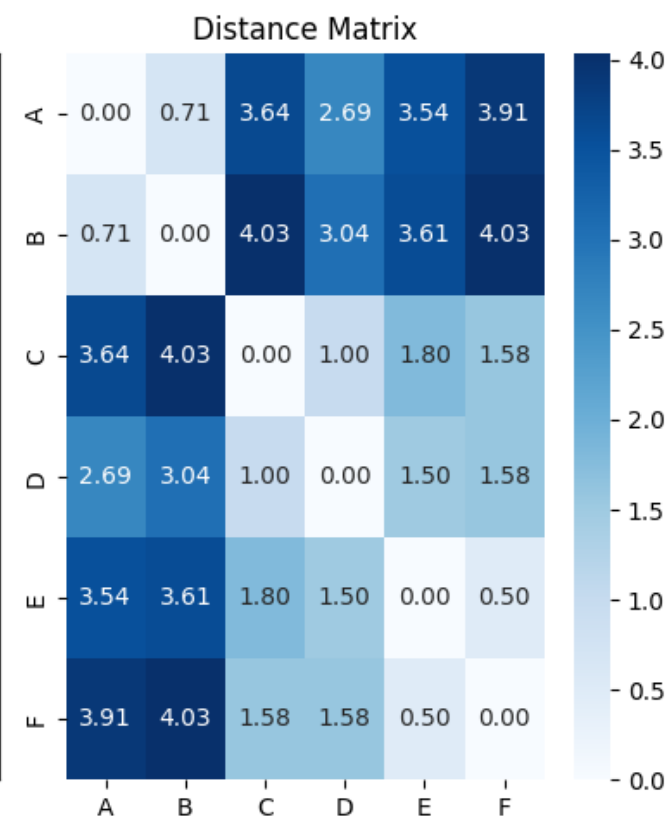
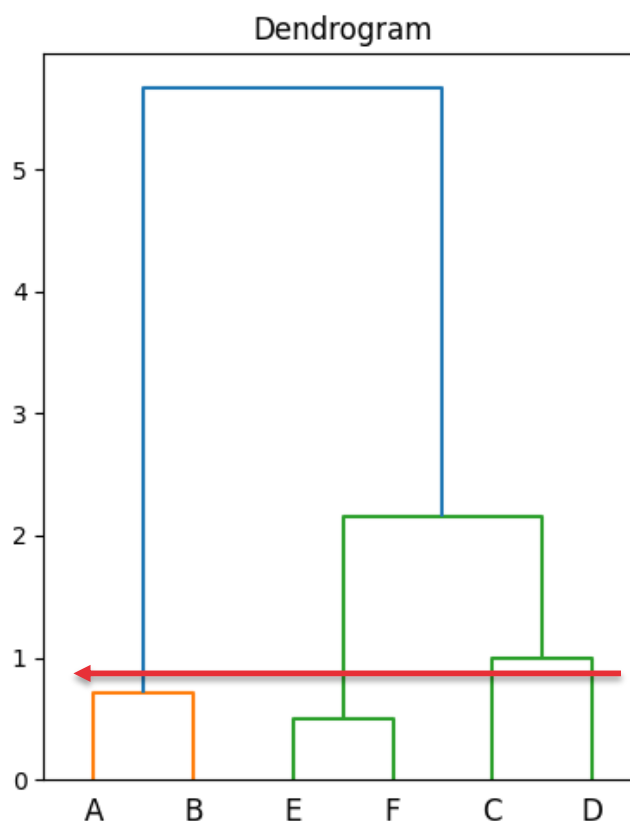
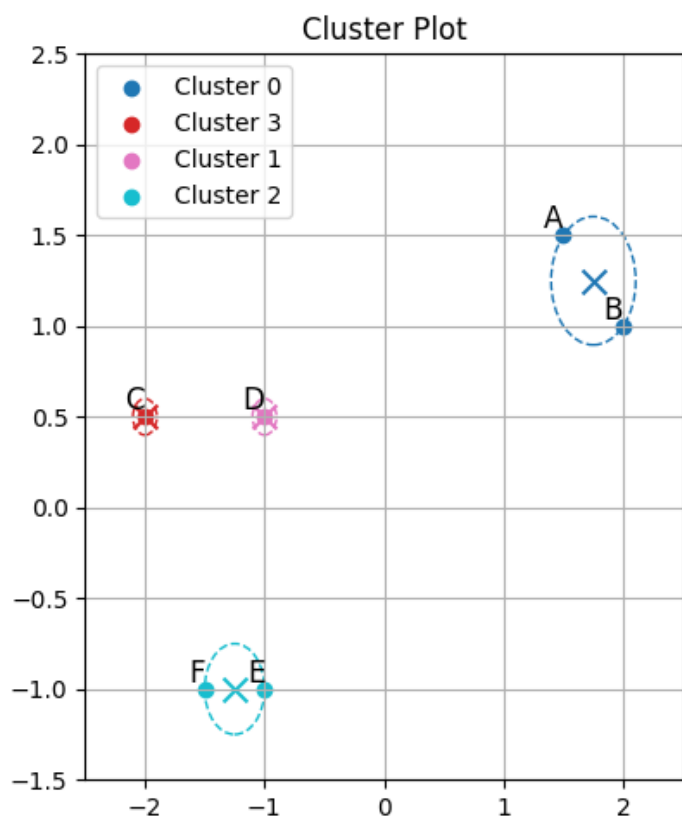
Single-linkage clustering

Discovering Groups – Agglomerative Clustering



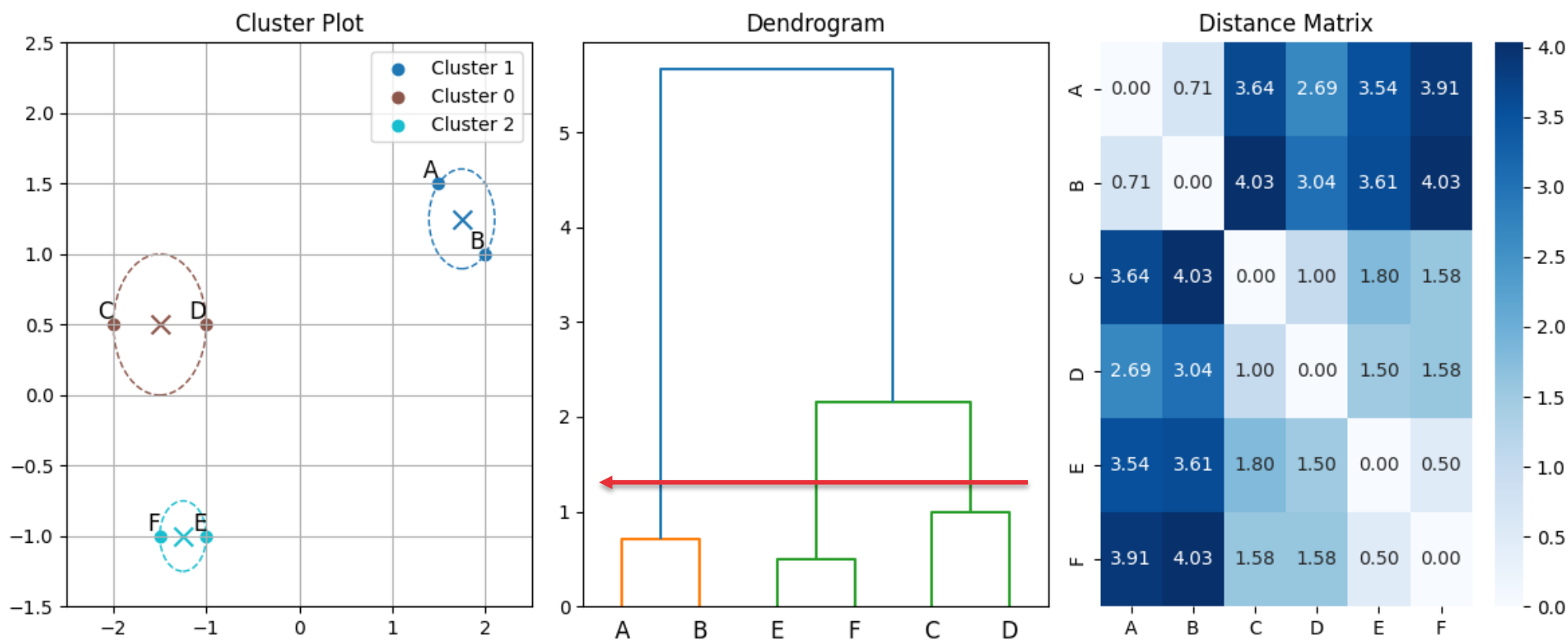
Single-linkage clustering

Discovering Groups – Agglomerative Clustering



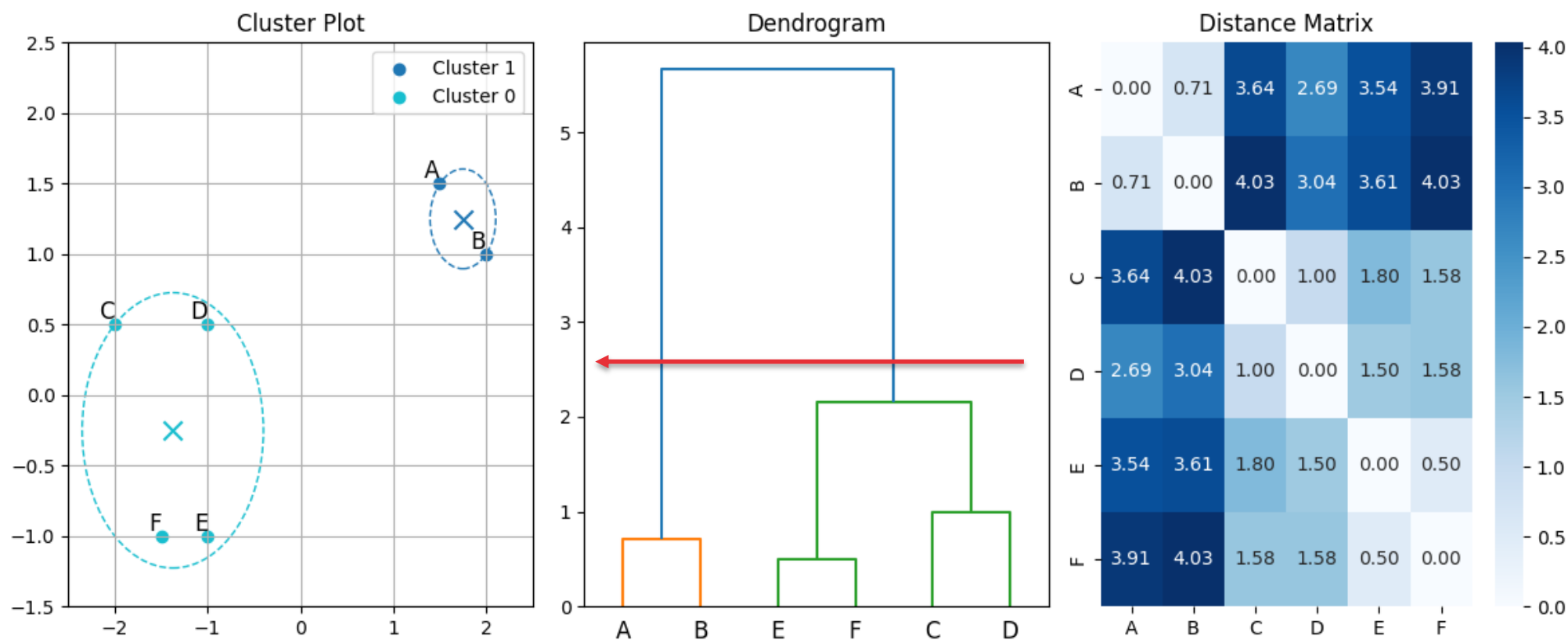
Single-linkage clustering

Discovering Groups – Agglomerative Clustering



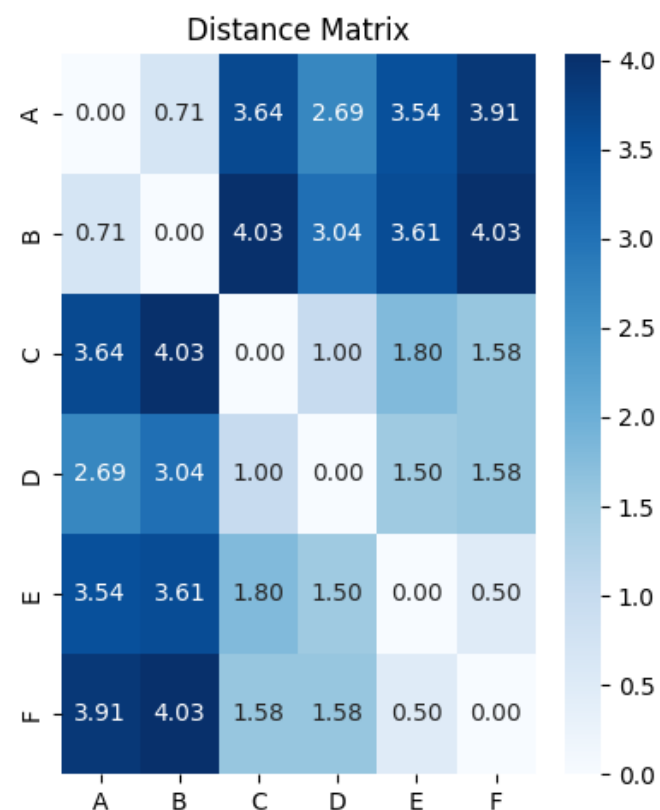
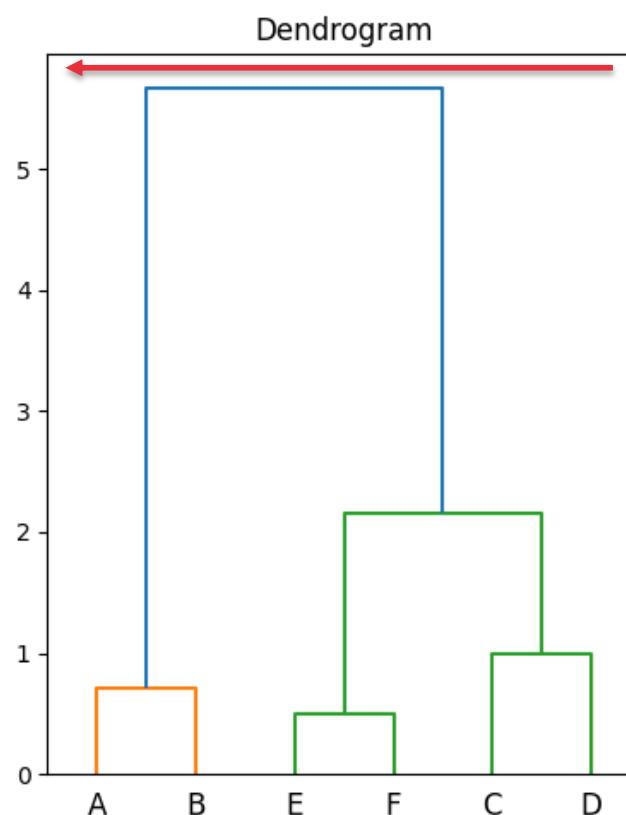
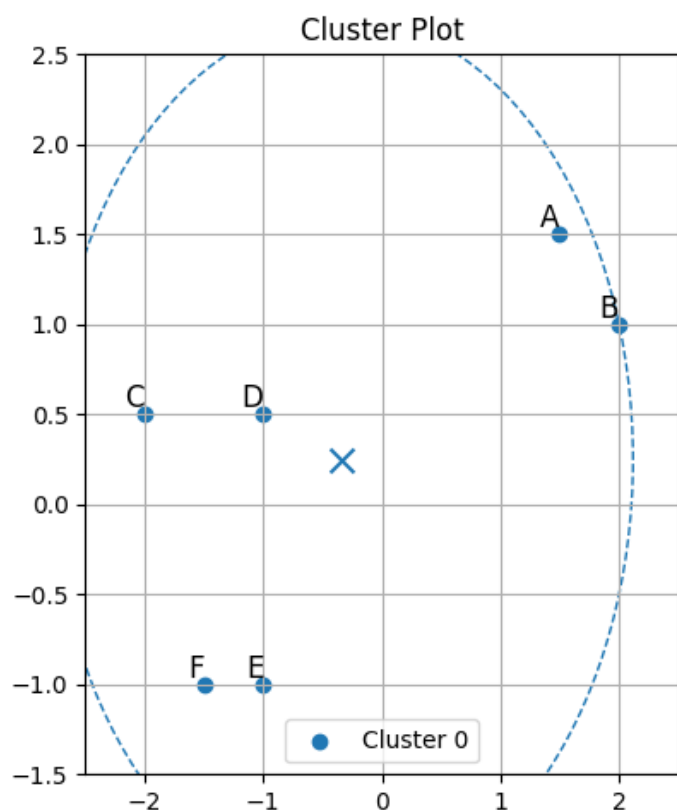
Single-linkage clustering

Discovering Groups – Agglomerative Clustering



Single-linkage clustering

Discovering Groups – Agglomerative Clustering



Single-linkage clustering

Discovering Groups – Hierarchical Clustering

Pros:

- No need to pre-specify cluster numbers; cut the dendrogram at the desired level for the clusters.
- Dendrograms easily summarize data into a hierarchy, facilitating cluster examination and interpretation.

Cons:

- Needs a threshold to determine the number of clusters
- Non-trivial to select the best linkage method

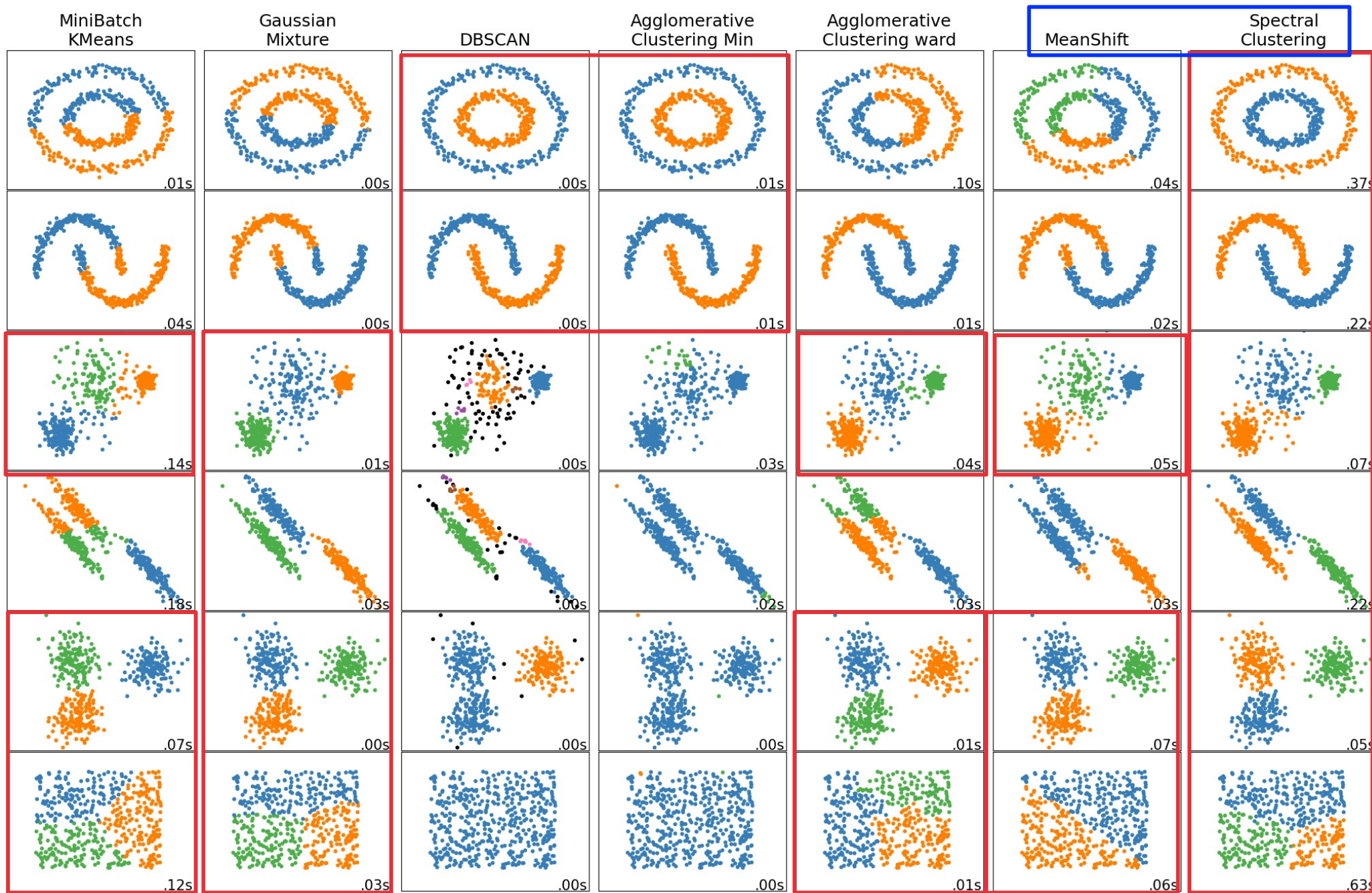
Discovering Groups – Summary

Clustering is a key way to understand your data.

There are many different approaches

- ▶ K Means - Need to chose K
- ▶ Hierarchical Agglomerative Clustering - needs a threshold or number of clusters

They are a very good way to start exploring a dataset



Source: https://scikit-learn.org/stable/auto_examples/cluster/plot_cluster_comparison.html#sphx-glr-auto-examples-cluster-plot-cluster-comparison-py

Discovering Groups – Learning Outcomes

- **LO1:** Comprehend the key ideas and the essential mathematical formulations employed in clustering methods (exam).
 - ❖ E.g., how is sum of squared error (SSE) defined?
 - ❖ E.g., understand the pros and cons of the learned algorithms
- **LO2:** Compute the fundamental stages of learned clustering approaches (exam).
 - ❖ E.g., given a dataset and a distance metric, be prepared to follow the selected clustering algorithm to cluster the instances in the dataset
- **LO3:** Implement and evaluate the learned clustering algorithms using Python (coursework)

Assessment hints: Multi-choice Questions (single answer: concepts, calculation etc)

- *Textbook Exercises: textbooks (Programming + Mining)*
- *Other Exercises: <https://www-users.cse.umn.edu/~kumar001/dmbook/sol.pdf>*
- *ChatGPT or other AI-based techs*

Discovering Groups – Textbook

CHAPTER 3

Discovering Groups

- ▶ Programming Collective Intelligence: Building Smart Web 2.0 Applications *T. Segaran*.

Chapter 7

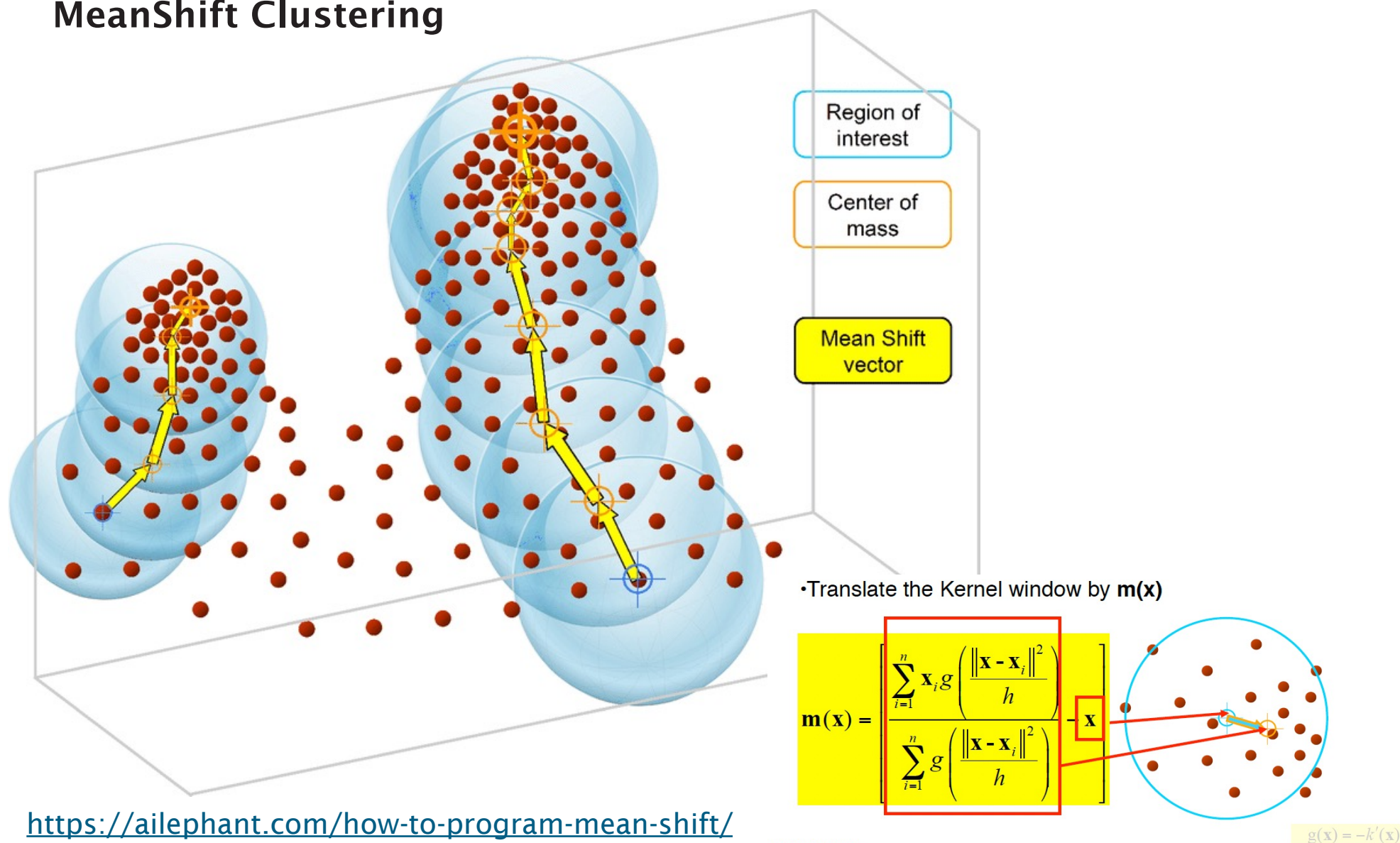
Clustering

Clustering is the process of examining a collection of “points,” and grouping the points into “clusters” according to some distance measure. The goal is that points in the same cluster have a small distance from one another, while points in different clusters are at a large distance from one another. A suggestion of

- ▶ Mining of Massive Datasets J. Leskovec *et al*

Discovering Groups – Appendix (1/2)

MeanShift Clustering



<https://ailephant.com/how-to-program-mean-shift/>

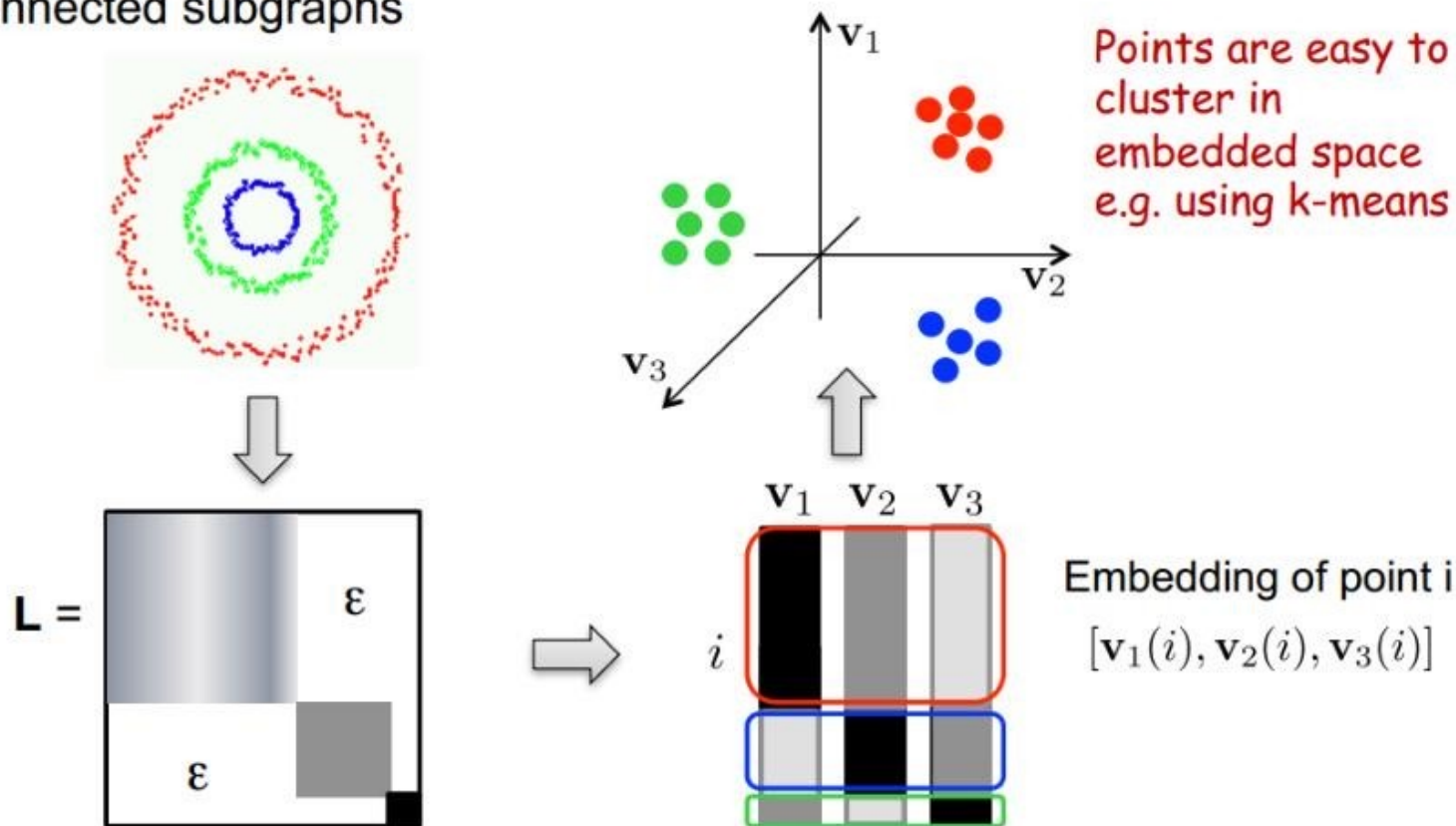
Source: Savarese slides.

Discovering Groups – Appendix (2/2)

Spectral Clustering

Eigenvectors of the Laplacian matrix provide an embedding of the data based on similarity.

Disconnected subgraphs



Slides Courtesy: Eric Xing, M. Hein & U.V. Luxburg